Groupware Systems & KM 2.0

Groupware is a term that refers to technology designed to help people collaborate and includes a wide range of applications. Wikipedia defines three handy categories for groupware:

- **Communication tools:** Tools for sending messages and files, including email, webpublishing, wikis, filesharing, etc.
- **Conferencing tools:** e.g. video/audio conferencing, chat, forums, etc.
- **Collaborative management tools:** Tools for managing group activities, e.g. project management systems, workflow systems, information management systems, etc.

The best known groupware system is Lotus Notes.

If designed and implemented properly, groupware systems are very useful when it comes to supporting knowledge management (KM). They can greatly facilitate explicit knowledge sharing through publishing and communication tools. They can support the knowledge creation process with collaborative management tools - although this process is still very much about people interacting and experimenting. Finally, they have some limited benefit to tacit knowledge transfer by supporting socialization through tools like video conferencing and informal communication. Expert finders are also beneficial for facilitating the location of tacit sources of knowledge (i.e. the right expert).

Web 2.0, Enterprise 2.0, & KM 2.0

In recent years, the term web 2.0 has appeared to describe the increasingly popular tools that promote two way communication on the internet. These social tools include blogs, wikis, social bookmarking, commenting, shared workspaces, micro blogging and polling (Bebensee et al. 2010). They differ from traditional publishing in that they “put the knowledge sharing power in the hands of the users themselves” (Gurteen, 2012).

The web 2.0 tools that have been applied within organizations have been called enterprise 2.0 (Bebensee et al., 2010), and even more recently, the mapping of these principles to KM has been dubbed KM 2.0 (Gurteen, 2012).

It is fair to say that KM 2.0 is very much in its early stages. For this reason, I will discuss it separately from the very general groupware subsections below.

The value of KM 2.0: According to Cronk (2011, p. 84), web 2.0 tools “facilitate the development of social capital through knowledge sharing, which in turn increases the potential to create intellectual capital.” The author defines social capital as the total resources existing across social networks. Wright et al. (2010) find that the adoption of social media is not being championed by KM to the extent that one might expect, representing a missed opportunity.

Limitations of KM 2.0: Enterprise & KM 2.0 systems can suffer from the same failure factors as other KMS (more on this in the sections below). A failure example is presented by Garcia-Perez & Ayres (2009), who outlined the failure of an enterprise wiki. The study found that time needed to access & contribute to the wiki as well as the achievement of critical mass (i.e. having an adequate ratio of contributors) were failure factors. Furthermore, the authors warn that just because when asked employees claim

that they will share knowledge, that does not actually mean they will do so when the system is implemented.

Considerations for Groupware Acquisition/Design

When acquiring or developing groupware it is important to establish the functions that best match the organization's needs (further considerations for IT implementation can be found in the knowledge management system subsection). Remember, these are not solutions, they are at most enablers of KM.
Determine the processes that take place in the organization as well as how knowledge is currently stored and distributed, and establish how certain functions would improve them. **Focus on the informal** - both in terms of communication as well as the organizational networks and communities - so as to enhance rather than stifle creativity and innovation, and to increase the probability of acceptance.
Cheah (2007) points to the fact that many off the shelf groupware solutions could be greatly improved. For instance, email clients are designed in such a way so as to focus on the current email, but generally have limited functions for drawing knowledge and information from past mails. Document management systems also can be limited in their ability to extract knowledge from old documents. In both cases he suggests improved utilization of user-input metadata (including keywords that are weighed by importance) and categorization mechanisms, which would allow for effective knowledge mining in the future.
The point I am making by including this is that all groupware is not equal, and one should consider the functions from the perspective of long-term knowledge reuse (when applicable). For this to be possible, the intermediary knowledge packaging/sanitizing/categorizing process must be done in light of future expectations and requirements.

Groupware Implementation Issues

Groupware implies that workers are willing and able to work together and to share their knowledge. Implementation of groupware systems have had numerous failures. These have been attributed to:

"lack of top executive support, the proliferation of incompatible collaborative tools, installation of inadequate tools, end-user confusion, and existing work practices that are designed around individual rather than collaborative work" (Jones 2005 in Janson et al 2008).

Janson et al (2008) highlight that the success of groupware implementation hinges on the success that the organization has had in developing a culture where collaboration and sharing are the norm. Without that, there is no incentive to use the systems to their full potential.

The issue of standardization should also be considered. Will the systems be used in the same way and with the same rules throughout the organization? This should not be enforced if it may lead to a lack of acceptance and/or if it is not practically feasible.

Selection of the groupware systems should be influenced by the users or in the very least be carried out by someone who is knowledgeable in both the functions of the system and the work practices of the users. Too often, this is not the case, and the decision is made at the top levels of the organization, by individuals who hardly ever operate the system (Grudin 2003 in Janson 2008).

Consider also how the chosen tools relate to one another, and if one should invest in an integrated solution. For instance, an integrated groupware system composed of many complementary modules may be easier on the user since it implies getting accustomed to one brand. The same systems manufactured by different companies could be far harder to learn. At the same time, they may not offer as many features individually, and this too must be carefully balanced against the firm's specific needs.

Advantages and disadvantages of knowledge management

Knowledge management is a systematic approach to capturing and making use of a business' collective expertise to create value. The potential advantages of effective knowledge management are significant but, as with most processes, there are certain challenges to consider.

Advantages of knowledge management

Some of the common **benefits of knowledge management** include:

- improved organisational agility
- better and faster decision making
- quicker problem-solving
- increased rate of innovation
- supported employee growth and development
- sharing of specialist expertise
- better communication
- improved business processes

A good **knowledge management system** will make it easy to find and reuse relevant information and resources across your business. This, in turn, can help you to:

- create better products and services
- develop better strategies
- improve profitability
- reuse existing skills and expertise
- increase operational efficiency and staff productivity
- recognise market trends early and gain an advantage over your rivals
- benchmark against your competitors
- make the most of your collective intellectual capital

Resourceful collaboration will bring more views, diverse opinions and varied experiences to the process of decision-making, helping your business to make decisions based on collective knowledge and expertise.

Disadvantages of knowledge management

The key to any successful knowledge management system is knowing its limitations. Some of the **common challenges** include:

- finding ways to efficiently capture and record business knowledge
- making information and resources easier to find
- motivating people to share, reuse and apply knowledge consistently
- aligning knowledge management with the overall goals and business strategy
- choosing and implementing knowledge management technology
- integrating knowledge management into existing processes and information systems

To overcome these challenges, you should:

- develop clear processes to capture, record and share business knowledge
- define the scope and objectives of any knowledge management initiatives
- create a corporate culture of knowledge sharing between employees and management
- set clear goals and strategies to help you utilise the collective knowledge (otherwise, it will be of no use to your business)
- consider budget, strategy and training needs for any new knowledge management system
- consider change management strategies for introducing new knowledge management practices

KM Systems

The issue of knowledge management systems has probably always been the most discussed and debated topic within knowledge management (KM). However, in modern KM, few people would disagree with the notion that knowledge management systems are an absolutely critical part of a KM initiative. More details on implementing this here: Knowledge Management Systems.
On this site, I have considered the impact of IT in all the knowledge management strategy subsections, with particular emphasis on its role in knowledge sharing. From this point on, the discussion will be organized as follows:

- This subsection will discuss the theoretical implementation of knowledge management systems and its impact on the organization.
- The section titled "KM Tools" will look at some of the main categories of systems available.

Looking for a Knowledge Management solution?

What are Knowledge Management Systems?

Knowledge management systems refer to any kind of IT system that stores and retrieves knowledge, improves collaboration, locates knowledge sources, mines repositories for hidden knowledge, captures and uses knowledge, or in some other way enhances the KM process.

If my explanation above makes the definition of these systems seem vague, that is because there is no consensus as to what constitutes a knowledge management system, much like there is no consensus regarding KM. Furthermore, since KM is involved in all areas of the firm, drawing a line is very difficult.

James Robertson (2007) goes as far as to argue that organizations should not even think in terms of knowledge management systems. He argues that KM, though enhanced by technology, is not a technology discipline, and thinking in terms of knowledge management systems leads to expectations of "silver bullet" solutions. Instead, the focus should be determining the functionality of the IT systems that are required for the specific activities and initiatives within the firm. However, with proper implementation, IT systems have become a critical component of KM today.

For the purpose of this site (intended to be useful for those people that do search for terms like knowledge management systems), I will break these down into the following general categories (adapted from the work of Gupta and Sharma 2005, in Bali et al 2009):

- Groupware systems & KM 2.0
- The intranet and extranet
- Data warehousing, data mining, & OLAP
- Decision Support Systems
- Content management systems
- Document management systems
- Artificial intelligence tools
- Simulation tools
- Semantic networks

These categories will cover the vast majority of the systems that people would normally associate with a KM system.

Problems and Failure Factors

Too often, the effects of technology on the organization are not given enough thought prior to the introduction of a new system. There are two sets of knowledge necessary for the design and implementation of a knowledge management system (Newell et al., 2000):

- The technical programming and design know-how
- Organizational know-how based on the understanding of knowledge flows

The problem is that rarely are both these sets of knowledge known by a single person. Moreover, technology is rarely designed by the people who use it. Therefore, firms are faced with the issue of fit between IT systems and organizational practices, as well as with acceptance within organizational culture (Gamble & Blackwell 2001).

Botha et al (2008) stress the importance of understanding what knowledge management systems cannot do. They point to the fact that introducing knowledge sharing technologies does not mean that experts will share knowledge - other initiatives have to be in place.

Akhavan et al (2005) identify several additional failure factors including: lack of top management support, organizational culture, lack of a separate budget, and resistance to change.

Building upon all this, and incorporating previously discussed elements, failure factors of knowledge management systems are as follows:

- Inadequate support: managerial and technical, during both implementation and use.
- Expecting that the technology is a KM solution in itself.
- Failure to understand exactly what the firm needs (whether technologically or otherwise).
- Not understanding the specific function and limitation of each individual system.

- Lack of organizational acceptance, and assuming that if you build it, they will come – lack of appropriate organizational culture.
- Inadequate quality measures (e.g. lack of content management).
- Lack of organizational/departmental/etc fit - does it make working in the organization. easier? Is a system appropriate in one area of the firm but not another? Does it actually disrupt existing processes?
- Lack of understanding of knowledge dynamics and the inherent difficulty in transferring tacit knowledge with IT based systems (see segment on tacit knowledge under knowledge sharing).
- Lack of a separate budget.

Promoting Acceptance and Assimilation

According to Hecht et al. (2011) the process of successful implementation has three stages: adoption, acceptance, and assimilation. Based on recognized models and theories, the authors identified three comprehensive sets of factors affecting these three elements. The resulting model organized the KMS implementation factors into the following categories:

- Adoption:
- Influenced by design: Innovation characteristics, fit, expected results, communication characteristics.
- Not influenced by design: Environment, technological infrastructure, resources, organizational characteristics.
- Acceptance
- Influenced by design: Effort expectancy, performance expectancy.
- Not influenced by design: Social influences, attitude towards technology use.
- Assimilation:
- Influenced by design: social system characteristics, process characteristics.
- Not influenced by design: Management characteristics, institutional characteristics.

Step 1: KMS Adoption

Some of the key factors identified by Hecht et al (2011) are: characteristics, commercial advantage, cultural values, information quality, organizational viability, and system quality. To promote KMS adoption:

- Start with an internal analysis of the firm.
- Evaluate information/knowledge needs & flows, lines of communication, communities of practice, etc. These findings should form the basis of determining the systems needed to complement them.

- Make a thorough cost-benefit analysis, considering factors like size of firm, number of users, complexity of the system structure, frequency of use, upkeep & updating costs, security issues, training costs (including ensuring acceptance) etc. vs improvements in performance, lower response time, lower costs (relative to the previous systems) etc.
- Evaluate existing work practices and determine how the systems will improve - and not hinder - the status quo.
- One very interesting rule of thumb presented by Botha et al (2008), is that "the more tacit the knowledge, the less high-tech the required solution". For example, expert knowledge is often best supported by multimedia communication technology and by expert finders. Beyond that, it is about human interaction and collaboration.

Step 2: KMS acceptance

Some of the factors outlined by Hecht et al. (2011) include: anxiety, ease of use, intrinsic motivation, job-fit, results demonstrability, and social factors. Promoting acceptance can be improved by:

- Involve the users in the design and implementation process when possible (Liebowitz 1999).
- Involve the user in the evaluation of the system when applicable (Liebowitz 1999).
- Make it as user friendly and as intuitive as possible (Frank 2002).
- Support multiple perspectives of the stored knowledge (Frank 2002).
- Provide adequate technical and managerial support.
- Use product champions to promote the new systems throughout the organization.

Step 3: KMS Assimilation

Some of the factors identified by Hecht et al. (2011) include: knowledge barrier, management championship, process cost, process quality, and promotion of collaboration. Assimilation can be improved by:

- Content management (Gamble & Blackwell, 2001): In order for the system to remain useful, its content must be kept relevant through updating, revising, filtering, organization, etc.
- Perceived attractiveness factors (Gamble & Blackwell, 2001): This includes not only the advantages of using the KMS, but also of management's ability to convince users of these advantages.
- Proper budgeting: i.e. planning expenses and implementing a KMS that is cost efficient.
- Focus on collaboration. In particular, consider the adoption of enterprise 2.0 / KM 2.0 systems, which by design promote collaboration while generally being inexpensive and often quite popular.
- Management involvement: The system must be championed by management at all levels.

Naturally, these factors do not apply to all systems. Some are fairly straightforward and accepted in today's society (e.g. email). However, the strategic implications of implementing knowledge management systems that significantly aim to change the way things are done in the organization requires proper consideration and careful planning. Moreover, with the evolution of systems to better support different facets of KM, they should be regarded as a critical component in the implementation of the discipline.

Content Management Systems

Content management systems are very relevant to knowledge management (KM) since they are responsible for the creation, management, and distribution of content on the intranet, extranet, or a website. Content management is a discipline in itself, so this section will be relatively brief, only outlining the basic considerations.

A content management system may have the following functions:

- Provide templates for publishing: Making publishing easier and more consistent with existing structure/design.
- Tag content with metadata: I.e. Allowing the input of data that classifies content (e.g. keywords) so that it can be searched for and retrieved.
- Make it easy to edit content
- Version control: Tracking changes to pages and, if necessary, allowing previous versions to be accessed
- Allow for collaborative work on content
- Integrated document management systems
- Workflow management: Allowing for parallel content development
- Provide extensions and plug-ins for increased functionality
- Etc.

Content management systems come in different forms (and prices), and an organization must carefully evaluate what it needs. Tanya Sahu (2007) presents six general factors for consideration:

- **Technology:** Including dynamic vs static publishing, high load performance, security issues, and search engine ranking factors (static pages rank better).
- **Ease of use:** Most users are non-technical. Therefore, it is important to assess the ease of use of the end user content editing interface, the template-building interface, and the content approval system.
- **Total cost of ownership:** I.e. the costs in the long run, including maintenance and applications.
- **Cross Platform Support and Scalability:** Can it handle multiple operating systems? Can it integrate with other server side technologies?
- **Web Presence Management:** The system should allow for the management of different websites separately and securely so as to manage multiple web presences (e.g. site on the intranet vs site on the extranet).
- **Solution deployment:** How long will it take to move content onto the content management system and how hard will the process be?

James Robertson (2003) stresses that the processes that surround the content management system are of most value to the management of knowledge. Apart from what has been already discussed, he emphasizes processes such as the restructuring

and rewriting of content carried out by professional writers supported by experts. This not only improves the accessibility and presentation, but also points to content gaps.

As one can see, selection and implementation of a content management system is something that requires careful consideration. As with all KM related IT systems, the functionality must be weighed against organizational needs and processes as well as expected costs. If properly implemented, the content management system can be very beneficial to KM, by improving the quality of explicit knowledge, and providing limited support to tacit knowledge transfer by identifying content authors (i.e. experts) and supporting collaborative projects.

Document Management Systems

Document management systems, as the name implies, are systems that aid in the publishing, storage, indexing, and retrieval of documents. Although such systems deal almost exclusively with explicit knowledge, the sheer volume of documents that an organization has to deal with makes them useful and in some cases even mandatory. Often they are a part of content management systems.

Usually, a document management system will include the following functions:

- **Capturing:** In order for paper documents to be useable by the document management system, they must be scanned in. For companies that need to carry out this process and who have numerous paper documents this may be time consuming and expensive.
- **Classification using metadata:** Metadata (data about data) is used to identify the document so that it can be retrieved later. It can include keywords, date, author, etc. The user is often asked to input this metadata or the system may extract it from the document. Optical character recognition may be used to identify text on scanned images.
- **Indexing:** There are many different forms, and a good indexing system is crucial. The index function will use metadata.
- **Searching & retrieval:** The document management system's search function is one of its most important elements. Search functions can be more or less sophisticated, allowing for searches by elements of the document's metadata, or by searching the actual document for key words/phrases and using semantic analysis to determine relevance.
- **Versioning:** Storage and management of different versions of documents - useful for documents that require frequent updating. Allows authorized users to return to earlier versions.
- **Administration & security:** Any IT system needs to be regulated and policed. Users require different levels of authorization, with certain more sensitive functions/documents being available only to selected users/administrators. Document management systems will also have backup systems in place in case of mishaps.

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Beyond this, they may include a whole host of possible features, including multiple platform support, multiple/customizable interfaces, workflow modules, file/format conversions, etc. Prices vary accordingly and solutions should be carefully matched to organizational requirements. There is also an open source system called OpenKM that, despite its somewhat misleading use of the term "knowledge management", seems to have gathered a significant degree of popularity.

Compared to non-electronic systems, the document management systems offer reduced operational costs, improved efficiency and speed of retrieval, improved consistency, and more safety (both in terms of file backups and security measures). There are hundreds of systems out there, each with their own strengths and weaknesses. The points to consider are: purchasing/set up costs, types of features, training, upkeep, ease of use etc. Be sure to assess each one in light of your specific needs and your organizational processes.

Knowledge Management: Processes, Techniques, And Tools

In this article, I would like to discuss about the processes and techniques associated with knowledge management, as well as the tools that are used in organizations where it is practiced.

Processes, Techniques, And Tools Of Knowledge Management

In the last installment, we looked at knowledge management, its constituent components, and the tasks it sets out to accomplish. In this article, I would like to tell you about the processes and techniques associated with knowledge management. In addition, we will take a look at tools that are commonly used in organizations where knowledge management is practiced and valued.

Processes

To ensure that knowledge in your company is properly created, stored, and used, it is paramount to implement the internal processes the right way. Company policy and procedures govern the way employees interact with each other and the organization

itself, so if you are looking to adopt knowledge management both effectively and efficiently, here are some things you should know:

1. General Company Policy Towards Knowledge Management.

Communicate the importance of knowledge management and knowledge sharing to your employees. Adopt knowledge management practices and make them mandatory in the day-to-day operations of your company. For example, giving employees some time off work every week so that they may think of what they had learned recently and commit it to writing is a good policy you may wish to implement.

2. Knowledge Creation Guidelines.

Make sure that any material that enters your knowledge base does so in an organized fashion. Create guidelines for knowledge creation, communicate them to your employees, and make sure that they are followed. Ideally, every employee should have access to the knowledge management tools and be encouraged to contribute.

However, keep in mind that the guidelines must help employees create and store knowledge, not stifle their enthusiasm. You will do well to create a template for adding new material to the knowledge base. The template should make it easy to understand what information should be displayed on every page, how the material should be structured, what fonts are to be used, how the text should be formatted, and so on. This will make life much easier for those who are willing to put in the time to make knowledge management a success in your organization while at the same time ensuring that the information in your knowledge base is well-structured and consistent. In these parts, we call this a “win-win”.

3. Contacting The Subject Matter Expert.

In the previous article, we established that it is necessary to give your employees access to the foremost experts within your company. Establishing rules that tell how and when the experts may be consulted will, on the one hand, give your employees access to superior expertise, and on the other, make sure that the experts can perform their main duties without constant interruption. Best practice in this regard is to have every expert pick a day of the week and set some time aside for consulting at that day at a specific time (make sure to communicate this information to your employees). Also, it is recommended for those seeking advice to clearly formulate their questions beforehand to save time.

4. Preserving Expertise Related To Solved Issues.

Companies, like people, must learn from their errors and avoid pitfalls they've encountered before. To that end, it is vital to preserve the information about the incident, as well as the steps taken to rectify it. You will be wise to make it a rule in your organization that as soon as an issue has been dealt with, the following data must be entered into your knowledge base:

- What was the issue.
- What caused the issue.
- How the issue was solved (if it was solved).
- The contact information of the employee(s) who dealt with the issue so that them may be reached at short notice should the issue arise again.

5. Preserving The Expertise Of Employees Leaving The Company.

Employee turnover is a fact of life, and even key personnel may leave for greener pastures. However, this doesn't mean that you can't or shouldn't try and minimize the damage this will cause. Expertise is valuable, and you should strive to preserve every ounce of it once somebody gives you their notice. While frontal learning will come in handy, you should try and have the leaving person's knowledge committed to writing as well as communicated orally. If you are parting ways on good terms, the soon to be ex-employee will probably not refuse to create a document detailing the processes they were responsible for. In addition, ask them to highlight the potential pitfalls that may cause issues unless handled with care. Preserving the expertise of departing employees is crucial for any organization, so take it seriously, or risk taking a big loss.

Knowledge Management Tools And Techniques

In knowledge management, technology plays second fiddle to people and the interactions between them. However, tools and techniques also have their place. The platform for implementing knowledge management that you choose must feature fundamental knowledge management tools and be user friendly and easy to operate. This is important - if you make participating in knowledge management a chore, don't be surprised when your employees are less than enthused by the prospect. To understand what features an ideal knowledge management platform must possess, let us take a look at the tools commonly used for that purpose in companies around the world.

Knowledge Bases

The “old reliable” option, knowledge management adoption in a company often starts with creating a knowledge base (to be fair, it often doesn’t progress any further than that). It usually takes the form of an intranet portal or a wiki-based mini-site containing answers to the most frequently asked questions in the organization, descriptions of technical procedures, detailed use cases and post-mortems of issues encountered in the past. A knowledge base can contain text, multimedia materials, video tutorials, links to relevant third party resources, and more.

Ideally, the knowledge base should be maintained and updated regularly. At a minimum, you should have a few employees tasked with compiling new expertise within the company and entering it in the knowledge base in an agreed upon format. Preferably, every employee should have the ability to edit the knowledge base and add new material to it.

From the technology standpoint, there are a number of possible options for building a knowledge base, including in-house solutions, wiki software, and LMS/LCMS (stands for Learning Management System and Learning Content Management System, respectively), with the latter being the preferable option. Unlike websites and wikis, LMS/LCMS feature the functionality necessary for knowledge management, from uploading material and training courses to collecting usage statistics, right out of the box.

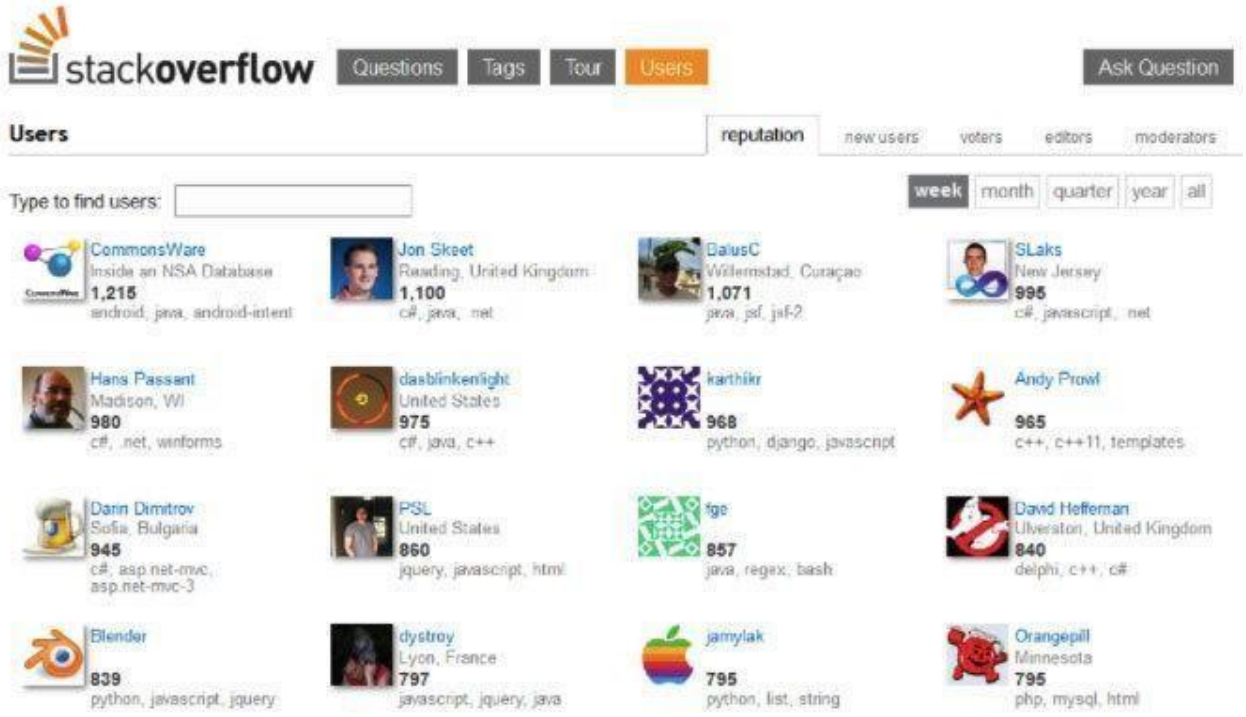
Communities Of Practice

These are professional communities where members are united by a common sphere of activity, a shared project, or a common goal. If you can make it so that your employees start talking shop with each other, sharing tips and experience, congratulate yourself on getting over a major knowledge management hurdle, as creating and fostering such communities is one of the major knowledge management goals. You can use different technical means, such as forums or groups belonging to the organization's internal social network, to help your employees share their expertise. It would also be a big plus if the chosen solution features the ability to rate or upvote particularly useful topics or insightful contributions.

A self-regulating community is a treasure trove of useful ideas and innovative solutions. The ability to have a quick informal chat with a colleague raises the employees' confidence and makes them more efficient workers. In most cases, leaving a post on a forum or shooting a private message on a social network is quicker and more convenient than conversing via email.

Knowledge Map

A knowledge map makes it easy for your employees to connect to an expert in a specific area of knowledge. It stands to reason that before you can ask for advice, you must first find someone who can give a competent answer, and this is where a knowledge map comes in handy. It lists the experts within the organization, shows their respective areas of expertise, and provides the means of contacting them.



Bank Of Ideas

Bank of ideas collects and stores the ideas and best practices generated in your organization. Companies known for their groundbreaking innovation don't simply get their brilliant ideas out of thin air - they actively solicit ideas from their employees and managers, and they listen. However, not every idea can be immediately considered for implementation, so in the meantime you need to store them somewhere. You also need to record who came up with what, so that when the company decides to act upon a particular suggestion, its author can be consulted and rewarded. As always, having the ability to rate other people's ideas and comment upon them is a huge plus.

No outside consultant, however many years they may have spent in the industry, knows your company as intimately as the employees who've spent years working in it. Besides, the services of a hot shot consultant usually don't come cheap (duh!). Give your employees a voice, make it clear that it will be heard, and who knows - your company may just become the next Apple or Google.

The Different Types of Knowledge

Understanding the different forms that knowledge can exist in, and thereby being able to distinguish between various types of knowledge, is an essential step for knowledge management (KM). For example, it should be fairly evident that the knowledge captured in a document would need to be managed (i.e. stored, retrieved, shared, changed, etc.) in a totally different way than that gathered over the years by an expert craftsman.

Over the centuries many attempts have been made to classify knowledge, and different fields have focused on different dimensions. This has resulted in numerous classifications and distinctions based in philosophy and even religion. Though not directly related to our purpose here, the wikipedia article on knowledge provides some interesting background reading.

Within business and KM, two types of knowledge are usually defined, namely explicit and tacit knowledge. The former refers to codified knowledge, such as that found in documents, while the latter refers to non codified and often personal/experience-based knowledge.

KM and organisational learning theory almost always take root in the interaction and relationship between these two types of knowledge. This concept has been introduced and developed by Nonaka in the 90's (e.g. Nonaka 1994) and remains a theoretical cornerstone of this discipline. Botha et al (2008) point out that tacit and explicit knowledge should be seen as a spectrum rather than as definitive points. Therefore in practice, all knowledge is a mixture of tacit and explicit elements rather than being one or the other. However, in order to understand knowledge, it is important to define these theoretical opposites.

Some researchers make a further distinction and talk of embedded knowledge. This way, one differentiates between knowledge embodied in people and that embedded in processes, organizational culture, routines, etc. (Horvath 2000). Gamble and Blackwell (2001) use a scale consisting of represented-embodied-embedded knowledge, where the first two closely match the explicit-tacit.

Without question, the most important distinction within KM is between explicit and tacit knowledge. However, I find that the embedded dimension is a valuable addition, since the managerial requirements for this type of knowledge are quite different. For this reason, the discussions on this site will, when relevant, use all three categorizations of knowledge but the focus will always be primarily on the explicit-tacit dimension.

Below I present an overview of these three categories, as well as a short discussion on the way knowledge management systems (KMS) can/cannot be used to manage them.

Explicit Knowledge

This type of knowledge is formalized and codified, and is sometimes referred to as know-what (Brown & Duguid 1998). It is therefore fairly easy to identify, store, and retrieve (Wellman 2009). This is the type of knowledge most easily handled by KMS, which are very effective at facilitating the storage, retrieval, and modification of documents and texts.

From a managerial perspective, the greatest challenge with explicit knowledge is similar to information. It involves ensuring that people have access to what they need; that important knowledge is stored; and that the knowledge is reviewed, updated, or discarded.

Many theoreticians regard explicit knowledge as being less important (e.g. Brown & Duguid 1991, Cook & Brown 1999, Bukowitz & Williams 1999, etc.). It is considered simpler in nature and cannot contain the rich experience based know-how that can generate lasting competitive advantage.

Although this is changing to some limited degree, KM initiatives driven by technology have often had the flaw of focusing almost exclusively on this type of knowledge. As discussed previously, in fields such as IT there is often a lack of a more sophisticated definition. This has therefore created many products labeled as KM systems, which in actual fact are/were nothing more than information and explicit knowledge management software.

Explicit knowledge is found in: databases, memos, notes, documents, etc. (Botha et al. 2008)

Tacit Knowledge

This type of knowledge was originally defined by Polanyi in 1966. It is sometimes referred to as know-how (Brown & Duguid 1998) and refers to intuitive, hard to define knowledge that is largely experience based. Because of this, tacit knowledge is often context dependent and personal in nature. It is hard to communicate and deeply rooted in action, commitment, and involvement (Nonaka 1994).

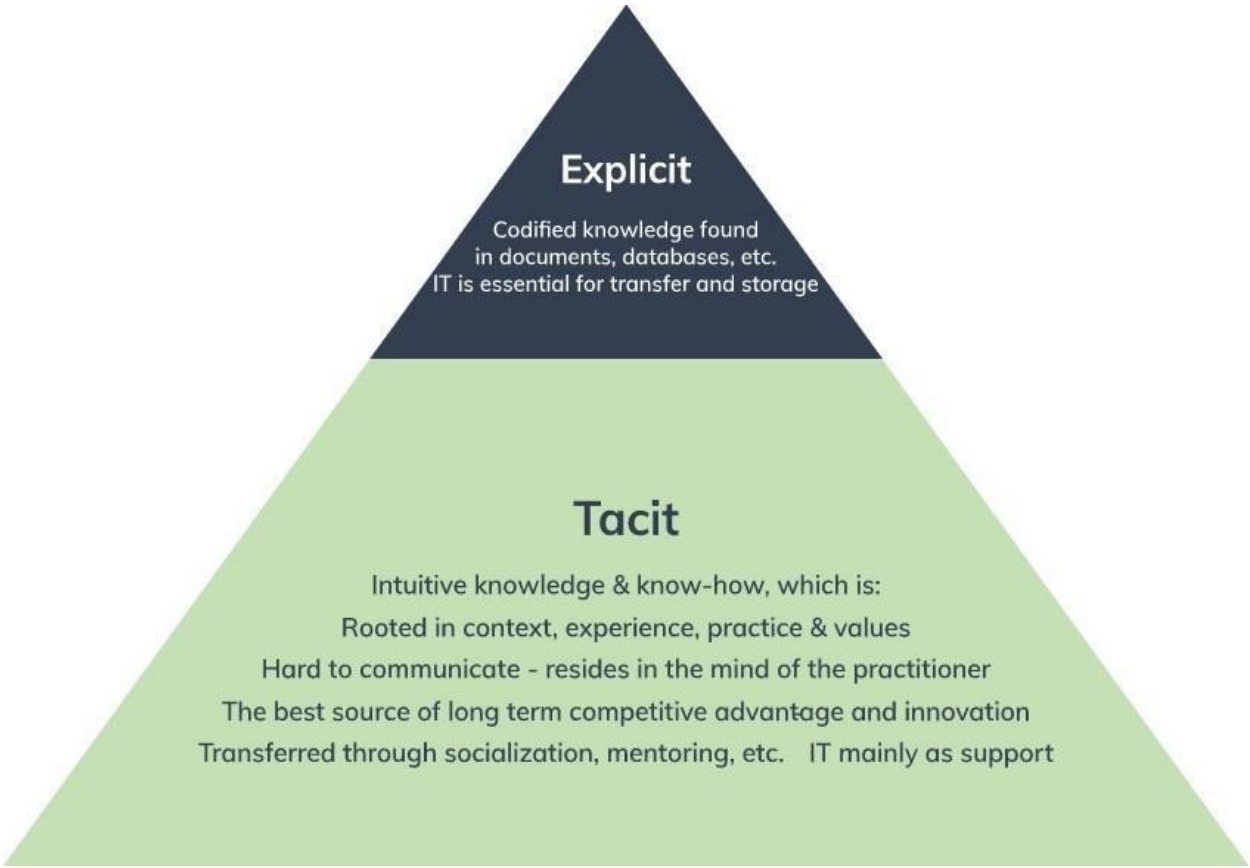
Tacit knowledge is also regarded as being the most valuable source of knowledge, and the most likely to lead to breakthroughs in the organization (Wellman 2009). Gamble & Blackwell (2001) link the lack of focus on tacit knowledge directly to the reduced capability for innovation and sustained competitiveness.

KMS have a very hard time handling this type of knowledge. An IT system relies on codification, which is something that is difficult/impossible for the tacit knowledge holder.

Using a reference by Polanyi (1966), imagine trying to write an article that would accurately convey how one reads facial expressions. It should be quite apparent that it would be near impossible to convey our intuitive understanding gathered from years of experience and practice. Virtually all practitioners rely on this type of knowledge. An IT specialist for example will troubleshoot a problem based on his experience and intuition. It would be very difficult for him to codify his knowledge into a document that could convey his know-how to a beginner. This is one reason why experience in a particular field is so highly regarded in the job market.

The exact extent to which IT systems can aid in the transfer and enhancement of tacit knowledge is a rather complicated discussion. For now, suffice it to say that successful KM initiatives must place a very strong emphasis on the tacit dimension, focusing on the people and processes involved, and using IT in a supporting role.

Tacit knowledge is found in: the minds of human stakeholders. It includes cultural beliefs, values, attitudes, mental models, etc. as well as skills, capabilities and expertise (Botha et al 2008). On this site, I will generally limit tacit knowledge to knowledge embodied in people, and refer separately to embedded knowledge (as defined below), whenever making this distinction is relevant.



Embedded Knowledge

Embedded knowledge refers to the knowledge that is locked in processes, products, culture, routines, artifacts, or structures (Horvath 2000, Gamble & Blackwell 2001). Knowledge is embedded either formally, such as through a management initiative to formalize a certain beneficial routine, or informally as the organization uses and applies the other two knowledge types.

The challenges in managing embedded knowledge vary considerably and will often differ from embodied tacit knowledge. Culture and routines can be both difficult to understand and hard to change. Formalized routines on the other hand may be easier to implement and management can actively try to embed the fruits of lessons learned directly into procedures, routines, and products.

IT's role in this context is somewhat limited but it does have some useful applications. Broadly speaking, IT can be used to help map organizational knowledge areas; as a tool in reverse engineering of products (thus trying to uncover hidden embedded knowledge); or as a supporting mechanism for processes and cultures. However, it has also been argued that IT can have a disruptive influence on culture and processes, particularly if implemented improperly.

Due to the difficulty in effectively managing embedded knowledge, firms that succeed may enjoy a significant competitive advantage.

Embedded knowledge is found in: rules, processes, manuals, organizational culture, codes of conduct, ethics, products, etc. It is important to note, that while embedded knowledge can exist in explicit sources (i.e. a rule can be written in a manual), the knowledge itself is not explicit, i.e. it is not immediately apparent why doing something this way is beneficial to the organization.

The Intranet & Extranet

The intranet is essentially a small-scale version of the internet, operating with similar functionality, but existing solely within the firm. Like the internet, the intranet uses network technologies such as Transmission Control Protocol/Internet Protocol (TCP/IP). It allows for the creation of internal networks with common internet applications that can allow them to communicate with different operating systems (Newell et al 2000).

Although it need not be, the intranet is usually linked to the internet, where broader searches are implemented. However, outsiders are excluded through security measures such as firewalls.

The Role of the Intranet

The intranet can be a very useful tool in the knowledge management process. It allows for the integration of multimedia communication and can act as a platform for groupware applications and publishing. It is intended to enhance collaboration, productivity, and socialization, but also to influence organizational culture and to act as a repository for embedded knowledge.

Robertson (2009) identifies seven key roles of the intranet homepage:

- news
- navigation
- key tools
- key information
- community and culture
- internal marketing
- collaboration

The focus is to provide a useful site that enhances work practices, communicates key information, provides the right navigation tools, and helps define organizational culture. Many factors have to be balanced to create the right homepage, including quality of content, site design, site navigation, site & content maintenance and updates, and the application of tools that are directly useful to the business processes and networks. The objectives of the intranet will also vary depending on the individual business, and may focus more on certain aspects than others.

Perhaps the most important function of the intranet is knowledge sharing and collaboration. The main functions supporting this are (Damsgaard & Scheepers 1998 in Newt et al 2000):

- **Publishing:** E.g. homepages, newsletters, documents, employee directories.
- **Searching:** The intranet can integrate different search functions, e.g. through a search engine or using a system of categorization.
- **Transacting:** Allows user to make transactions with other web/intranet homepages.

- **Interacting:** Collaborative applications and other groupware, expert finders, directories, etc.
- **Recording:** It can be used as a storage medium for such elements as procedures, best practices, and FAQs (embedded and explicit knowledge).

Successful Intranet Implementation

Naturally, the implementation of the intranet must be done in line with organizational needs, processes, and objectives, as outlined in the section on implementation of knowledge management systems.

One specific and key concern is the selection of the search engine. Google offers an option for on-site search, which you can read more about here.

In his article, "The Ten Best Intranets of 2011", Jakob Nielsen (2011) indicates that the best intranets implemented solutions in the following areas:

- **Knowledge sharing:** This aspect is very similar to what I have discussed so far on this site and includes the sharing of all manner of explicit knowledge, but also connecting people that require assistance to experts that can help them.
- **Innovation management:** By incorporating tools that support the recording and management of new ideas.
- **Comments:** This is an easy way to allow users to contribute with their insight. This type of loose, unstructured communication can provide some limited tacit knowledge transfer and can encourage participation.
- **Ratings:** An even quicker, albeit shallower, way for people to point to good sources of knowledge.
- **Participation rewards:** Point systems, badges, and other symbolic rewards actually increase participation. Sometimes non-symbolic rewards (i.e. actual prizes) were used.
- **Customized collections:** By allowing users to customize content collections, one can bypass the shortcoming of never being fully able to predict a user's knowledge and information needs.

The Extranet

The extranet is an extension of the intranet to the firm's external network, including partners, suppliers and so on. The term is sometimes used to refer to a supplementary system working alongside the intranet or to a part of the intranet that is made available to certain external users.

The extranet provides a shared network with limited, controlled access to organizational information and knowledge resources, and uses security protocols, such as authentification, to limit access. An extranet can enhance collaboration and information transfer with partners in the external network.

Security is a key concern, and a firm must protect its crucial knowledge and information resources. This can be done using firewalls, use of encryption, and simple or strong authentification. Simple authentification involves usernames and passwords, while strong authentification makes use of digital certificates.

The content of both intranets and extranets is usually managed with a content management system.

Warehousing Data: The Data Warehouse, Data Mining, and OLAP

Warehousing data is based on the premise that the quality of a manager's decisions is based, at least in part,on the quality of his information. The goal of storing data in a centralized system is thus to have the means to provide them with the right building blocks for sound information and knowledge. Data warehouses contain information ranging from measurements of performance to competitive intelligence (Tanler 1997).

Data mining tools and techniques can be used to search stored data for patterns that might lead to new insights. Furthermore, the data warehouse is usually the driver of data-driven decision support systems (DSS), discussed in the following subsection.

Thierauf (1999) describes the process of warehousing data, extraction, and distribution. First data extraction of operational production data takes place, and this data is passed on to the warehouse database. A server hosts the data warehouse and the DSS. This server then passes on the extracted data to the warehouse database, which is employed by users to extract data through some form of software.

Theirauf's model for data warehousing is as follows:

Warehousing Data: Design and Implementation

Tanler (1997) identifies three stages in the design and implementation of the data warehouse. The first stage is largely concerned with identifying the critical success factors of the enterprise, so as to determine the focus of the systems applied to the warehouse. The next step is to identify the information needs of the decision makers. This involves the specification of current information lacks and the stages of the decision-making process (i.e. the time taken to analyze data and arrive at a decision). Finally, warehousing data should be implemented in a way that ensures that users understand the benefit early on. The size of the database and the complexity of the analytical requirements must be determined. Deployment issues, such as how users will receive the information, how routine decisions must be automated, and how users with varying technical skills can access the data, must be addressed.

According to Frank (2002), the success of the implementation of the data warehouse depends on:

- Accurately specifying user information needs
- Implementing metadata: Metadata is essentially data about data. This is regarded as a particularly crucial step. Parankusham & Madupu (2006) outline the different roles of metadata as including: data characterization and indexing, the facilitation or restriction of data access, and the determination of the source and currency of data. They further identify the lifecycle of metadata as:
- Collection: Identification and capture
- Maintenance: Updating of metadata to match changes in data architecture

- Deployment: Users access the relevant metadata, based on their needs.

To this, we can add the 5 criteria presented on the www.syntelinc.com website:

- Recognize that the job is probably harder than you expect: A large portion of the data in data warehouses is incorrect, missing, or input in such a way that it is not usable (e.g. historical databases that have not been updated to modern schemas).
- Understand the data in your existing systems: Analyze existing databases. Identify relationships between existing data systems so as to avoid inconsistencies when these are moved to the warehouse.
- Be sure to recognize equivalent entities: Identify equivalent entities in heterogeneous systems, which may appear under a different name.
- Emphasize early wins to build support throughout the organization
- Consider outsourcing your data warehouse development and maintenance: Implementing a data warehouse can be a huge task that can often be better handled by experts. Many data warehousing applications are suited for outsourcing.

If properly designed and implemented, the goal of warehousing data is to drastically reduce the time required in the decision making process. To do so, it employs three tools, namely Online Analytical Processing System (OLAP), data mining, and data visualization (Parankusham & Madupu 2006).

OLAP

OLAP allows three functions to be carried out.

- Query and reporting: Ability to formulate queries without having to use the database programming language.
- Multidimensional analysis: The ability to carry out analyses from multiple perspectives. Tanler (1997) provides an example of a product analysis that can be then repeated for each market segment. This allows for quick comparison of data relationships from different areas (e.g. by location, time, etc.). This analysis can include customers, markets, products, and so on,
- Statistical analysis: This function attempts to reduce the large quantities of data into formulas that capture the answer to the query.

OLAP is basically responsible for telling the user what happened to the organization (Theirauf 1999). It thus enhances understanding reactively, using summarization of data and information.

What is Data Mining?

This is another process used to try to create useable knowledge or information from data warehousing. Data mining, unlike statistical analysis, does not start with a preconceived hypothesis about the data, and the technique is more suited for heterogeneous databases and date sets (Bali et al 2009). Karahoca and Ponce (2009)

describe data mining as "an important tool for the mission critical applications to minimize, filter, extract or transform large databases or datasets into summarized information and exploring hidden patterns in knowledge discovery (KD)." The knowledge discovery aspect is emphasized by Bali et al (2009), since the management of this new knowledge falls within the KM discipline.

It is beyond the scope of this site to offer an in-depth look at the data mining process. Instead, I will present a very brief overview, and point readers that are interested in the technical aspects towards free sources of information.

Very briefly, data mining employs a wide range of tools and systems, including symbolic methods and statistical analysis. According to Botha et al (2008), symbolic methods look for pattern primitives by using pattern description languages so as to find structure. Statistical methods on the other hand measure and plot important characteristics, which are then divided into classes and clusters.

Data mining is a very complex process with different process models. One is the CRoss-Industry Standard Process for Data Mining (or Crisp-DM). The process involves six steps (Maraban et al, in Karahoca & Ponce 2009):

Business understanding -> data understanding -> data preparation -> modeling -> evaluation -> deployment

For more on data mining see the book "Data Mining and Knowledge Discovery in Real Life Applications", edited by Ponce & Karahoca (2009), available for free from intechopen.com where numerous other potentially relevant resources can also be downloaded.

Data Visualization

This process involves representing data and information graphically so as to better communicate its content to the user. It is a way to make data patterns more visible, more accessible, easier to compare, and easier to communicate. Data visualization includes graphical interfaces, tables, graphs, images, 3D presentations, animation, and so on (Turban & Aaronson in Parankusham & Madupu 2006).

DSS are other tools used in conjunction with warehousing data. These are discussed in the following subsection.